Tzu-Hsiang (Sean) LIN

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EDUCATION

University of Michigan **Ann Arbor, MI, USA**
Master of Science in Mechanical Engineering (Robotics) *2025-2027*
Coursework: *Mathematics for Robotics (ROB501), Robotics System Lab (ROB550), Mobile Robot: Algorithm (ROB530)*
Research Lab: DASC Lab; Zoetic Robotics Lab; SAM Lab

University of Nottingham **Nottinghamshire, UK**
Bachelor of Engineering (Hons) in Mechanical Engineering *2021-2025*
GPA: 3.90/ 4.00 (Top1%)
Honors: Nottingham Global **Scholarship**, Dean's **Scholarship,** Ningbo MoE **Scholarship**, Ningbo Government **Scholarship**

SKILLS

Programming & Software: C++, Python, C, MATLAB, Git, Linux, REST APIs, frontend/backend development, LLM
Robotics & Perception: ROS2, ROS, SLAM, OpenCV, YOLO, PyTorch, LiDAR, RGB-D, IMU, Kalman Filtering, Particle filter
Planning, Control & Optimization: A*, PID, LQR, MPC, CBF-MPC, MPPI, BR-MPPI, trajectory planning, Optuna
Simulation & Hardware: Gazebo, RViz2, NVIDIA Isaac Sim, PX4, Sim2Real, Simulink, Raspberry Pi, STM32, Arduino, ESP32
Mechanical: CAD (SolidWorks, CATIA, NX) FEA, 3D-printing, CFD, GD&T, DFM, 3D modelling, engineering drawing, machining

WORK EXPERIENCE

Apple Inc.
Product Design Engineering Intern *May. 2026- Sep. 2026*

- Supported Apple Watch New Product Introduction (NPI) builds, **FA** on design and manufacturing issues through hardware validation, tolerance analysis, and conducted **DOEs** during build and production ramp.
- Developed a full-stack **LLM-powered** engineering tool**,** contributing to frontend and backend development and integrating generative-AI capabilities through API calls and an internal GenAI integration layer**. Achieving time reduction on task by 87%**
- Implemented **backend** workflows for LLM inference, prompt/data routing, **API communication**, and structured response processing**,** enabling automated retrieval and analysis of engineering information.
- Developed a **computer-vision-based** dimensional analysis pipeline using object detection and image processing to localize component features and automatically calculate physical dimensions from images.
- Designed and implemented **a control algorithm** for water-seal testing fixture**, reducing test cycle time by 91%.**

Bellwether Industries
Mechanical Engineer Intern *Aug. 2024- Oct. 2024*

- Designed and integrated mechanical, propulsion, and onboard electronic systems for a 1/8-scale eVTOL prototype, supporting end-to-end vehicle development and testing.
- Built a multi-orientation flight-dynamics test platform for evaluating vehicle behaviour and propulsion
- Developed simulation and flight-system using MATLAB, Simulink, and PX4, and investigated MRAC for eVTOL
- Performed CFD and FEA to evaluate aerodynamic flow, structural loading, and design performance under multi-directional operating conditions.

PROJECT EXPERIENCE

TEG- SLAM (Temporal-Guided 3D Gaussian SLAM)/ University of Michigan **Ann Arbor, MI**
SLAM Engineer *Oct. 2025 - Present*

- Contributed to the development of a 3D Gaussian SLAM pipeline**,** extending Wild-GS SLAM with temporal information to improve trajectory estimation and scene reconstruction.
- Implemented a memory-aware pose estimation module by modifying Lie-group-based state updates and temporal feature integration within the SLAM optimization pipeline.
- Achieved an 18% reduction in ATE, improving reconstruction quality by 2–5% in PSNR and SSIM over the baseline**.**

Mobile Robot Perception, Navigation and Control/ Robotics System Lab / University of Michigan **Ann Arbor, MI**
Robotics Engineer *Oct. 2025 - Present*

- Developed a C++ environment for a differential-drive mobile robot using ROS2 on a Pi with 2D LiDAR, IMU, and wheel encoders for perception and designed cascade PID controllers for accurate motion control under uncertainty.
- Developed Bresenham's algorithm, Monte Carlo Localization for SLAM, and A* for autonomous navigation.
- Analyzed particle-filter SLAM runtime, achieving real-time localization at 10 Hz with up to thousands of particles.

5-DoF Robotics Arm Autonomy / Robotics System Lab / University of Michigan **Ann Arbor, MI**
Robotics Engineer *Sep. 2025 - Nov. 2025*

- Integrated computer vision sensing and kinematic reasoning for a 5-DOF robotic arm with an Intel RealSense camera.
- Applied intrinsic and extrinsic calibration for precise vision alignment and enabled semi-autonomous pick-and-place
- Coding Python file block detection via OpenCV and Pox-based kinematics, achieving 1.9% average positional error.

RESEARCH EXPERIENCE

University of Michigan / Distributed Autonomous Systems and Control Lab **Ann Arbor, MI**
Robotics Researcher *Jan. 2026 – Present*

- Developed and deployed **safety-critical control algorithms** for multi-agent aerial robots, integrating CBF-MPC and gatekeeper-based safety mechanisms for collision-constrained exploration.
- Implemented and validated control and autonomy modules on **Crazyflie hardware**, supporting both single-robot and multi-agent experiments in GPS-denied environments.
- Integrated planning, control, and motion-capture systems through custom **ROS interfaces and coordinate-frame transformations**, enabling closed-loop execution from simulation to physical hardware.
- Investigated real-world failure modes including controller instability, state-estimation error, communication latency, and model mismatch during close-proximity multi-robot flight.
- Built an Optuna-based hyperparameter tuning and benchmarking pipeline for BR-MPPI, optimizing controller parameters and evaluating safety, trajectory-tracking performance, and computational efficiency.

University of Michigan / Zoetic Robotics Lab **Ann Arbor, MI**
Robotics Researcher *Sep. 2025 – Present*

- Developed experimental instrumentation and data-acquisition workflows for pneumatically actuated bio-inspired aerial robots, involving in designing, simulation and mathematical modelling.
- Built image-processing and tracking workflows to quantify small actuator displacements from high-speed video and characterize dynamic response across operating conditions.
- Developed physics-based and semi-empirical models of nonlinear actuator behaviour and compared simulation predictions against experimental measurements.

University of Michigan / Synergetic Adaptive Machina's Lab **Ann Arbor, MI**
Robotics Researcher *Sep. 2025 – Present*

- Developing the high-level controller to change the magnetic field, manipulating the microrobot locomotion
- Setup experimental environment and design the circuits for magnetic field controlling

University of Nottingham / Advanced Robotics Laboratory **Nottingham, UK**
Mechanical (Robotics) Researcher *Aug. 2024 – Jun. 2025*

- Developed bio-inspired manta ray locomotion system; implemented kinematic model (PCC) for undulation/oscillation control in MATLAB. Use FEA for complex dynamic behavior under pressure from 30kPa to 100kPa to by ABAQUS.
- Fabricated and tested soft pneumatic actuators to validate simulation results, achieving a feasible robot prototype design

PUBLICATION

- S. Kim*, Y.-H. Hsiao*, A. Yin*, **T.-H. Lin**, J. Brown, G. Matt, Y. Chen, C. Aubin, **Controlled hovering flight in a sub-gram micro aerial vehicle powered by pneumatic micro-muscles** *(in prep)* (2025)
- T. Kim*, R. H. Kumar*, A. D. Menon, **T.-H. Lin**, D. Panagou, **SEAMLiS: Visibility-Aware Safety for Perception-Limited Multi-Robot Exploration**, *arXiv:2607.09959 [cs.RO]* (2026)